CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sphere and hemisphere

1. The volumes of two spheres are in the ratio $27 : 64$. Find the ratio of their radii. (a) $3 : 4$ (b) $4 : 3$ (c) $9 : 16$ (d) $16 : 9$ ICSE 2025, 1 mark

SOLUTION

Volume is proportional to r^3: $\frac{r_1^3}{r_2^3} = \frac{27}{64}$, so $\frac{r_1}{r_2} = \frac{3}{4}$.

Answer: (a)

2 Melting and recasting

2. A hollow sphere has an external diameter of 10 cm and an internal diameter of 6 cm. It is melted and made into a solid cone of height 8 cm. Find the radius of the cone. ICSE 2025, 3 marks

SOLUTION

Volume of metal $= \frac{4}{3}\pi(R^3 - r^3) = \frac{4}{3}\pi(5^3 - 3^3) = \frac{4}{3}\pi \times 98$

Volume of the cone $= \frac{1}{3}\pi x^2 \times 8$, where x is its radius.

$\frac{8}{3}\pi x^2 = \frac{4}{3}\pi \times 98$, so $x^2 = 49$.

Answer: 7 cm

3. A solid sphere of radius 6 cm is melted and recast into 64 identical spherical marbles. Find the radius of each marble. ICSE 2020, 3 marks

SOLUTION

$64 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6)^3$, so $r^3 = \frac{216}{64} = \frac{27}{8}$

Answer: $r = 1.5$ **cm**

4. A solid metal sphere of radius 6 cm is melted and made into a solid cylinder of height 32 cm. Find (a) the radius of the cylinder (b) its curved surface area. Take $\pi = 3.1$. ICSE 2019, 4 marks

SOLUTION

(a) $\pi r^2 \times 32 = \frac{4}{3}\pi(6)^3 = 288\pi$, so $r^2 = 9$ and $r = 3$ cm.

(b) Curved surface area $= 2\pi rh = 2 \times 3.1 \times 3 \times 32$

Answer: (a) 3 cm (b) 595.2 cm²

3 Combination of solids

5. A solid wooden capsule is a cylinder with a hemisphere on each end. Its radius is 3.5 cm and the cylindrical part is 14 cm long. The capsule is cut into three separate solids: the cylinder and the two hemispheres. Find the sum of the total surface areas of the three pieces. Take $\pi = \frac{22}{7}$.

ICSE 2025, 4 marks

SOLUTION

Each piece is now a separate solid, so its flat faces count as well.

Cylinder: $2\pi r(h + r) = 2 \times \frac{22}{7} \times 3.5 \times (14 + 3.5) = 22 \times 17.5 = 385$ cm²

One solid hemisphere: $3\pi r^2 = 3 \times \frac{22}{7} \times 3.5 \times 3.5 = 115.5$ cm². Two hemispheres: 231 cm².

Sum $= 385 + 231$

Answer: 616 cm²

6. A spherical vessel of radius 28 cm is three-quarters full of oil. The oil is poured into a cylindrical vessel of radius 21 cm. Find the height of the oil in the cylinder, to the nearest centimetre. ICSE 2024, 4 marks

SOLUTION

Volume of oil $= \frac{3}{4} \times \frac{4}{3}\pi(28)^3 = \pi \times 21952$ cm³

In the cylinder: $\pi(21)^2 h = \pi \times 21952$, so $h = \frac{21952}{441} \approx 49.78$

Answer: about 50 cm

7. A solid is a hemisphere of radius 7 cm with a cone of height 4 cm standing on its flat face. It is lowered completely into water in a cylindrical container of radius 14 cm. Find the rise in the water level.

ICSE 2023, 4 marks

SOLUTION

Volume of the solid $= \frac{2}{3}\pi(7)^3 + \frac{1}{3}\pi(7)^2(4) = \frac{686\pi}{3} + \frac{196\pi}{3} = 294\pi$ cm³

The water rises by h where $\pi(14)^2 h = 294\pi$, so $h = \frac{294}{196}$.

Answer: 1.5 cm

8. A solid wooden cylinder is 28 cm high and 6 cm in diameter. A conical cavity of diameter 6 cm and height 10.5 cm is hollowed out of each end. Find the volume of the remaining solid. Take $\pi = \frac{22}{7}$.

ICSE 2020, 3 marks

SOLUTION

Volume of the cylinder $= \pi(3)^2(28) = 252\pi$

Volume of the two cones $= 2 \times \frac{1}{3}\pi(3)^2(10.5) = 63\pi$

Remaining volume $= 189\pi = 189 \times \frac{22}{7}$

Answer: 594 cm³

9. A hemispherical hole and a conical hole are scooped out of the two ends of a solid wooden cylinder of height 7 cm and radius 3 cm. The hemisphere and the cone both have radius 3 cm, and the cone is 3 cm high. Find the volume of the remaining solid to the nearest whole number. Take $\pi = \frac{22}{7}$.

ICSE 2019, 4 marks

SOLUTION

Cylinder: $\pi(3)^2(7) = 63\pi$. Hemisphere: $\frac{2}{3}\pi(3)^3 = 18\pi$. Cone: $\frac{1}{3}\pi(3)^2(3) = 9\pi$.

Remaining volume $= 63\pi - 18\pi - 9\pi = 36\pi = 36 \times \frac{22}{7} \approx 113.14$

Answer: 113 cm³

10. A solid is a cylinder with a hemisphere on one end and a cone on the other, all of radius 7 cm. The cylinder and the cone are each 4 cm high. Find the volume of the solid. ICSE 2018, 4 marks

SOLUTION

Cylinder: $\pi(7)^2(4) = 196\pi$. Hemisphere: $\frac{2}{3}\pi(7)^3 = \frac{686\pi}{3}$. Cone: $\frac{1}{3}\pi(7)^2(4) = \frac{196\pi}{3}$.

Total $= 196\pi + \frac{882\pi}{3} = 196\pi + 294\pi = 490\pi = 490 \times \frac{22}{7}$

Answer: 1540 cm³

4 Cylinder

11. A rectangular sheet 11 cm by 7 cm is rolled into a cylinder in two ways: once with height 7 cm and once with height 11 cm. Find the ratio of the curved surface areas. (a) $1:1$ (b) $7:11$ (c) $11:7$ (d) $\frac{11\pi}{7} : \frac{7\pi}{11}$

ICSE 2024, 1 mark

SOLUTION

In both cases the curved surface is the sheet itself, with area $11 \times 7 = 77$ cm².

Answer: (a)

12. A solid metal cylinder of diameter 7 cm and height 10 cm is cut into two equal halves along its height. Find (a) the total surface area of the two halves together (b) the cost of painting both halves at ₹30 per cm². Take $\pi = \frac{22}{7}$. ICSE 2024, 4 marks

SOLUTION

(a) The two halves have all the surface of the cylinder plus two new flat rectangular faces, each 7 cm by 10 cm.

Cylinder: $2\pi r(h + r) = 2 \times \frac{22}{7} \times 3.5 \times 13.5 = 297$ cm². Two rectangles: $2 \times 70 = 140$ cm².

Total $= 297 + 140 = 437$ cm²

(b) Cost $= 437 \times 30$

Answer: (a) 437 cm² (b) ₹13,110

13. A cylinder of height 3 cm has a volume of 48π cm³. Find its radius. (a) 48 cm (b) 16 cm (c) 4 cm (d) 24 cm ICSE 2023, 1 mark

SOLUTION

$\pi r^2 \times 3 = 48\pi$, so $r^2 = 16$ and $r = 4$ cm.

Answer: (c)

14. The base of a cylindrical vessel has a circumference of 132 cm, and the vessel is 25 cm high. Find (a) its radius (b) its volume. Take $\pi = \frac{22}{7}$. ICSE 2018, 3 marks

SOLUTION

(a) $2 \times \frac{22}{7} \times r = 132$, so $r = 21$ cm.

(b) Volume $= \frac{22}{7} \times 21 \times 21 \times 25$

Answer: (a) 21 cm (b) 34,650 cm³

5 Cone

15. A conical tent must hold 77 people, each needing 16 m³ of air. The radius of the tent is 7 m. Find its height and its curved surface area. ICSE 2017, 3 marks

SOLUTION

Volume needed $= 77 \times 16 = 1232$ m³

$\frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times h = 1232$, so $\frac{154}{3}h = 1232$ and $h = 24$ m.

Slant height $l = \sqrt{24^2 + 7^2} = 25$ m. Curved surface area $= \frac{22}{7} \times 7 \times 25 = 550$ m².

Answer: height 24 m; curved surface area 550 m²

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